Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/28/2007
PI Name:	Lahey, Richard Ph.D.		
Project Title:	Analysis of the Effect of Gravity on Multiphase Flows		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3590	**Congressional District:**	21
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	08/03/2004	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GB43G		
Performance Goal No.:			
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Task Description:	NASA has initiated an ambitious and exciting program for the human exploration and development of space. This program includes a proposed manned-mission to Mars which will be necessarily of relatively long duration, and require the on-board production and use of a significant amount of power for space craft propulsion, station keeping and life support. It is expected that a nuclear fusion reactor will be the primary on-board energy source and that phase change systems will be required for many energy conversion purposes. Unfortunately, very little is currently known about the performance of multiphase systems and processes in space, and virtually no detailed experimental data is available [Viskanta et al, 2000]. Significantly, the National Research Council has identified this technology as both critical and mission enabling for NASA. The proposed research is focused on performing direct numerical simulations (DNS) of various multiphase flows at reduced gravity levels. The resultant numerical data should yield important new insights into flow regime specific phenomena at reduced gravity levels and will provide the basis for the future development of the interfacial and wall		

	closure laws required for the computational multiphase fluid dynamic (CMFD) models recommended by the NRC study [Viskanta et al, 2000].
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Considerable progress has been made to date on the numerical simulation of gas/liquid multiphase flows in the absence of gravity. At the microscale level a disjoining pressure based model was used to account for the wall forces that induce a fully wetting fluid to wrap around the walls of its container in the absence of gravity. This approach has the important advantage that no correlations for the contact angle are needed and that an overwhelming amount of theoretical and experimental evidence supports the concept of a disjoint pressure. We initially simulated the transient of Carbon Tetrachloride (CCL) moving up a vertical glass wall after gravity suddenly becomes zero. Liquid velocities of the order of 1 m/s are reached in fractions of a second. At the macroscale level we present simulations of unstable capillary waves for an air/water system in zero gravity. To the best of our knowledge these are the first numerical simulations reported of unstable capillary waves for a density ratio of 1,000. By merging the microscale and macroscale solvers we developed a multiscale solver which was used to simulate the lateral flow in a horizontal cylindrical pipe for microgravity conditions. As a first step in benchmarking our multiscsale solver we show that the constants that characterize the liquid-wall pair in the disjoint pressure model can be obtained from the static measurements of contact angle conducted by Weisgosel & Ross (1990). In particular ,the time evolution of the interface reported in this work is being used to benchmark our multiscale code. We have also shown that we can accurately simulate the motion of Hexane driven by a moving vertical wall and that we can efficiently account for surface tension effects.
Bibliography Type:	Description: (Last Updated: 12/28/2007)
Articles in Peer-reviewed Journals	Lahey RT Jr. "The simulation of multidimensional multiphase flows." Nuclear Engineering and Design 2005 May;235(10-12):1043-60. http://dx.doi.org/10.1016/j.nucengdes.2005.02.020 , May-2005
Articles in Peer-reviewed Journals	Galimov AY, Drew DA, Lahey RT Jr, Moraga FJ. "The analysis of interfacial waves." Nuclear Engineering and Design 2005 May;235(10-12):1283-92. http://dx.doi.org/10.1016/j.nucengdes.2005.02.016 , May-2005
Articles in Peer-reviewed Journals	Nagrath S, Jansen KE, Lahey RT Jr. "Computation of incompressible bubble dynamics with a stabilized finite element level set method." Computer Methods in Applied Mechanics and Engineering, 2005 Oct 15;194(42-44):4565-87. http://dx.doi.org/10.1016/j.cma.2004.11.012 , Oct-2005